Meeting 9 – Part 1

Logical Data Flow Architecture

Project: Digital Harmonium Workstation

Version: Final Discussion Draft

Status: Finalized Architecture

9.1 Purpose

The previous meetings established:

- Runtime states
- Software modules
- Task architecture
- Memory architecture

This meeting answers a different question:

> **How does information move through the workstation?**

Every feature is implemented as one or more data pipelines.

A pipeline begins when an event occurs and ends when the event has completed.

Examples include:

- Pressing a key
- Selecting an instrument
- Starting tabla
- Recording
- Playing back
- Updating the display

The purpose of this document is to identify every major logical data flow inside the workstation.

Group A — Audio Generation

Flow 1 — Keyboard Performance

```
Keyboard
↓
Keyboard Scanner
↓
Key Event
↓
Voice Manager
↓
Envelope Engine
↓
Sample Engine
↓
Effects
↓
Audio Mixer
↓
Limiter
↓
DMA Buffer
↓
I²S
↓
DAC
↓
Amplifier
↓
Speaker
```

Purpose:

Primary sound generation pipeline.

This is the most frequently executed data flow in the entire workstation.

Flow 2 — Key Release

```
Key Release
↓
Voice Manager
↓
Release Envelope
```

```
↓
Audio Mixer
↓
DMA
↓
Speaker
```

Purpose:

Handles note release without allocating a new voice.

The envelope naturally fades the sound until the voice becomes inactive.

Flow 3 — Polyphonic Mixing

```
Multiple Voices
↓
Audio Mixer
↓
Limiter
↓
Output
```

Purpose:

Combines all simultaneously active voices into a single stereo audio stream while preventing clipping.

Group B — Instrument Loading

Flow 4 — Instrument Installation

```
Instrument Selection
↓
Instrument Manager
↓
Loader Task
↓
SD Card
↓
```

```
Read VHL
↓
Flash Installation
↓
Verification
↓
Ready
```

Purpose:

Installs the selected instrument into the dedicated Flash partition.

Flow 5 — Metadata Initialization

```
Flash
↓
Metadata Parser
↓
Runtime Tables
↓
Voice Manager
```

Purpose:

Loads lookup tables, key mappings, regions, and playback metadata required for performance.

Group C — Tabla

Flow 6 — Tabla Playback

```
Play Button
↓
Tabla Engine
↓
Beat Scheduler
↓
Sample Trigger
↓
Voice Manager
```

```
↓
Mixer
↓
Speaker
```

Purpose:

Generates rhythmic accompaniment according to the selected taal.

Flow 7 — Tabla Tempo

```
Tempo Button
↓
Tempo Manager
↓
Beat Interval
↓
Tabla Engine
```

Purpose:

Updates beat timing without interrupting playback.

Group D — Tanpura

Flow 8 — Tanpura Playback

```
Play Button
↓
Tanpura Engine
↓
Drone Scheduler
↓
Voice Manager
↓
Mixer
```

Purpose:

Produces continuous drone accompaniment.

Flow 9 — Tanpura Pitch

```
Transpose
↓
Pitch Manager
↓
Tanpura Engine
```

Purpose:

Adjusts drone pitch according to the musician's selected tonic.

Group E — Metronome

Flow 10 — Metronome Playback

```
Start
↓
Metronome Engine
↓
Tick Generator
↓
Click Sample
↓
Mixer
```

Purpose:

Produces timing reference clicks independently of tabla playback.

Group F — Recording & Playback

Flow 11 — Recording

```
Key Events
↓
Recorder
↓
Recorder Buffer
↓
Background Writer
↓
SD Card
```

Purpose:

Stores performance events safely without disturbing audio playback.

Flow 12 — Performance Playback

```
Recording File
↓
Playback Reader
↓
Virtual Key Events
↓
Voice Manager
↓
Mixer
```

Purpose:

Recreates the original performance by replaying key events rather than recorded audio.

Flow 13 — LED Playback

```
Playback Engine
↓
Current Note
↓
LED Manager
```

```
↓
Keyboard LEDs
```

Purpose:

Illuminates the corresponding keys during playback to provide an educational learning aid.

Group G — User Interface

Flow 14 — Menu Navigation

```
Button
↓
Button Manager
↓
UI Manager
↓
Current Screen
↓
LCD
```

Purpose:

Processes user navigation throughout the workstation interface.

Flow 15 — Volume Control

```
Volume Slider
↓
Mixer Gain
↓
Audio Output
```

Purpose:

Adjusts overall speaker and headphone volume.

Flow 16 — LCD Contrast

```
Contrast Slider
↓
Display Driver
↓
LCD
```

Purpose:

Adjusts display readability without affecting runtime behavior.

Group H — Settings

Flow 17 — Save Settings

```
Settings Menu
↓
Settings Manager
↓
Flash
```

Purpose:

Stores user preferences permanently.

Flow 18 — Restore Settings

```
Flash
↓
Settings Manager
↓
Runtime Configuration
```

Purpose:

Restores previous configuration during boot.

Group I — Diagnostics

Flow 19 — Boot Self-Test

```
Power ON
↓
Diagnostics
↓
LED Test
↓
Audio Test
↓
LCD Test
↓
Ready
```

Purpose:

Verifies basic hardware before entering normal operation.

Flow 20 — Runtime Error

```
Subsystem
↓
Diagnostic Manager
↓
LCD Warning
↓
Log
```

Purpose:

Reports recoverable runtime faults to the user.

Group J — Background Services

Flow 21 — Memory Monitoring

```
Background Task
↓
Heap Monitor
↓
Diagnostics
```

Purpose:

Continuously supervises available memory.

Flow 22 — Voice Cleanup

```
Finished Voices
↓
Voice Manager
↓
Voice Pool
```

Purpose:

Returns completed voices to the available pool.

Flow 23 — Resource Cleanup

```
Unused Objects
↓
Background Task
↓
Free Memory
```

Purpose:

Removes temporary runtime objects no longer required.

Group K — Audio Hardware

Flow 24 — Physical Audio Output

```
Mixer
↓
DMA Buffer
↓
DMA Controller
↓
I²S
↓
UDA1334A DAC
↓
PAM8403 Amplifier
↓
Speakers / Headphones
```

Purpose:

Transfers completed digital audio into the analog domain for listening.

Group L — System Shutdown

Flow 25 — Shutdown Sequence

```
Shutdown
↓
Save Settings
↓
Flush Recorder Buffer
↓
Close SD Files
↓
Power Down
```

Purpose:

Ensures no user data is lost before the system powers off.

9.2 Summary

The workstation contains twenty-five primary logical data flows.

These pipelines collectively describe how information travels through the entire system, from user interaction to final audio output.

Unlike software modules, which describe *who performs the work*, logical data flows describe *how information moves* between those modules.

Together, these flows define the operational behavior of the Digital Harmonium Workstation and serve as the foundation for all future software implementation and system integration.

End of Meeting 9 – Part 1: Logical Data Flow Architecture